

9.1.19.

An urn contains two blue balls (denoted B_1 and B_2) and three white balls (denoted W_1 , W_2 , and W_3). One ball is drawn, its color is recorded, and it is replaced in the urn. Then another ball is drawn and its color is recorded.

- a. Let B_1W_2 denote the outcome that the first ball drawn is B_1 and the second ball drawn is W_2 . Because the first ball is replaced before the second ball is drawn, the outcomes of the experiment are equally likely. List all 25 possible outcomes of the experiment.
- b. Consider the event that the first ball that is drawn is blue. List all outcomes in the event. What is the probability of the event?
- c. Consider the event that only white balls are drawn. List all outcomes in the event. What is the probability of the event?

a. Let B_1W_2 denote the outcome that the first ball drawn is B_1 and the second ball drawn is W_2 . Because the first ball is replaced before the second ball is drawn, the outcomes of the experiment are equally likely. List all 25 possible outcomes of the experiment.

Let S be the set and finite sample space containing all possible outcomes when drawing balls from the urn containing colored balls, then:

$$S = \{B_1W_1, B_1W_2, B_1W_3, B_1B_1, B_2B_2, \\ W_1B_1, W_2B_1, W_3B_1, B_1B_2, B_2B_1, \\ B_2W_1, B_2W_2, B_2W_3, W_1W_1, W_2W_2, \\ W_1B_2, W_2B_2, W_3B_2, W_3W_3, W_1W_2, \\ W_2W_1, W_1W_3, W_3W_1, W_2W_3, W_3W_2\}$$

Solution: a. S , as listed above.

b. Consider the event that the first ball that is drawn is blue. List all outcomes in the event. What is the probability of the event?

Let $E_0 \subseteq S$, where E_0 is the set containing all outcomes where the first ball drawn is blue. Then:

$$E_0 = \{B_1W_1, B_1W_2, B_1W_3, B_1B_1, B_2B_2, \\ B_1B_2, B_2B_1, B_2W_1, B_2W_2, B_2W_3\}$$

Let $P(E_0)$ be the probability of E_0 , $N(E_0)$ be the number of outcomes in E_0 , and $N(S)$ be the total number of outcomes in S , then by definition of equally likely probability formula:

$$\begin{aligned} N(E_0) &= 10 \\ N(S) &= 25 \\ P(E_0) &= \frac{N(E_0)}{N(S)} = \frac{10}{25} = \frac{2}{5} = 0.4 \end{aligned}$$

Hence, the probability of the first ball drawn being blue is 40%.

Solution: b. The probability of the first ball drawn being blue is 40%.

c. Consider the event that only white balls are drawn. List all outcomes in the event. What is the probability of the event?

Let $E_1 \subseteq S$, where E_1 is the set containing all outcomes where only white balls are drawn. Then:

$$\begin{aligned} E_1 = \{ &W_1W_1, W_2W_2, W_3W_3 \\ &W_1W_2, W_2W_1, W_1W_3, \\ &W_3W_1, W_2W_3, W_3W_2 \} \end{aligned}$$

Let $P(E_1)$ be the probability of E_1 , $N(E_1)$ be the number of outcomes in E_1 , and $N(S)$ be the total number of outcomes in S , then by definition of equally likely probability formula:

$$\begin{aligned} N(E_1) &= 9 \\ N(S) &= 25 \\ P(E_1) &= \frac{N(E_1)}{N(S)} = \frac{9}{25} = 0.36 \end{aligned}$$

Hence, the probability of only drawing white balls is 36%.

Solution: c. The probability of only drawing white balls is 36%.